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TaskHive: A Web-Based Academic Project Management System for Higher Education

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Abstract
Effective management of student projects remains a significant challenge in higher education, often hampered by fragmented communication, poor task coordination, and insufficient oversight. This study introduces *TaskHive*, a web-based academic project management system tailored to the unique needs of higher education institutions. Designed to streamline project workflows, *TaskHive* enhances collaboration, task organisation, and real-time communication among students and supervisors. The system supports multiple user roles, students, supervisors, and team leads, each with dedicated access to features such as task assignment, document sharing, feedback provision, and performance tracking. Key modules include task management, document upload, integrated messaging, and a comprehensive evaluation dashboard that facilitates continuous monitoring of individual and group progress. By centralising all project-related activities within a single digital platform, *TaskHive* addresses common inefficiencies such as missed deadlines, lack of accountability, and supervision gaps. This research contributes to the digital transformation of academic administration by providing a scalable, intuitive solution that not only improves the execution of academic projects but also cultivates essential project management competencies among students, preparing them for real-world professional environments.

Keywords
Academic Project Management, Web-Based System, Higher Education, Collaboration Tools, Student Performance Tracking, Digital Transformation.

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Introduction

The rapid advancement of information and communication technologies has significantly transformed how individuals interact, learn, and work. From online learning and e-banking to digital shopping and virtual collaboration, these innovations have reshaped the educational landscape. However, academic project management, a critical component of higher education, has often lagged in embracing these technological advancements (Fowler et al., 2015; Singun, 2025). Traditional methods of managing student projects are typically manual, involving physical meetings, paper-based documentation, and basic communication tools such as emails or phone calls (Jääskä & Aaltonen, 2022). These approaches are time-consuming, prone to miscommunication, and often result in disorganised workflows and missed deadlines, thereby hindering project success (Sarkar et al., 2018). While semi-automated tools such as shared documents or generic task trackers provide some level of structure, they are not fully equipped to meet the specific needs of academic project teams (Mayer et al., 2024). These systems often lack the comprehensive features necessary for monitoring student progress, encouraging collaboration, and supporting real-time communication. In professional environments, project management software is often too complex or feature-rich for student use, creating a gap in accessible and practical solutions tailored to educational settings (Mas et al., 2020).

In student-led projects, leadership is typically delegated to a coordinator or team leader who oversees task distribution, team dynamics, and communication with external stakeholders such as lecturers. The effectiveness of such projects largely depends on the team's ability to manage time, meet quality expectations, and communicate effectively. A centralized platform that fosters transparency, accountability, and collaboration is crucial for reducing the risk of failure and enhancing academic outcomes.

This study proposes the design and implementation of *TaskHive*, a Web-Based Academic Project Management System tailored for higher education institutions. The system addresses existing challenges by offering a user-friendly interface that supports task management, document submission, real-time collaboration, status tracking, and supervisor feedback. By streamlining project workflows and enhancing team coordination, *TaskHive* aims to transform the student project experience, promote digital literacy, and contribute to a more structured, efficient, and collaborative academic environment.

Literature Review

Theoretical Foundation

The development of *TaskHive* is grounded in several key theories of project management and educational technology. One of the foundational models is the Project Management Body of Knowledge (PMBOK) (Morris, 2001), which outlines core project phases—initiation, planning, execution, monitoring, and closure. These PMBOK stages inform the functional design of project workflows within the

TaskHive system. It serves as a standard reference to define the essential topics and processes involved in managing projects effectively. The PMBOK has been developed and refined by leading professional organizations such as PMI and APM to underpin certification programs, guide practitioners' practices, and promote consistency across the industry. Additionally, Constructivist Learning Theory (Bada & Olusegun, 2015), which emphasises collaboration, student autonomy, and active engagement, supports the inclusion of interactive features that foster peer-to-peer and student-supervisor communication. It advocates for creating learning environments that encourage students to work together through interactive features, thereby fostering peer-to-peer communication. It supports meaningful interactions between students and supervisors, enhancing the overall learning experience by promoting reflection, shared expertise, and collective knowledge construction.

The Technology Acceptance Model (TAM) (Davis, 1989) also serves as a framework for evaluating user adoption of the platform, focusing on perceived usefulness and ease of use as critical factors in system implementation within educational contexts. The Technology Acceptance Model (TAM) serves as a framework for evaluating user adoption of a platform. It explains why users accept or reject information systems and how system design features influence user acceptance (Marangunić & Granić, 2015). It helps to identify key factors such as perceived usefulness (the degree to which a user believes the system enhances job performance) and perceived ease of use (the degree to which a user believes using the system requires minimal effort). These factors impact users' attitudes toward using a system or platform, which in turn influence actual usage behaviour. Additionally, it highlights the causal relationships between system design features, user perceptions, attitudes, and usage, making it a valuable tool for understanding and improving user adoption of technology platforms.

Academic Project Management Tools

Several digital tools exist to facilitate project management, such as Microsoft Teams[1], Trello[2], and Asana[3]. While effective in professional settings, these platforms often lack educational context-specific features, such as grading modules, supervisor feedback loops, and institutional integration. Studies such as Aithal and Aithal (2023) argue that the absence of tailored academic functions in such tools leads to limited student engagement and system underutilisation in higher education settings. Moreover, many tools are too complex or require extensive onboarding, posing a challenge for students unfamiliar with enterprise-level software. A project management tool, REDCap was developed by Garasević et al. (2024) to make managing clinical research. It improved research coordination and transparency by creating a centralised, digital research registry that streamlined data management and communication among stakeholders. Specifically, it enabled real-time monitoring of research activities, facilitated efficient communication via online surveys and input masks, and reduced administrative delays, such as the time to finalize clinical trial agreements. The digital platform replaced paper-based

[1] https://www.microsoft.com/en-us/microsoft-teams/group-chat-software
[2] https://trello.com/
[3] https://asana.com/

processes, allowing departments like laboratories and pharmacies to better track services, share essential documents promptly, and collaborate more transparently. Pan and Zhang (2021) implemented an advanced project management tool to facilitate project management as a data-driven digital twin framework that integrates Building Information Modeling (BIM), Internet of Things (IoT), and data mining (DM) techniques. This tool supports automation, digitalisation, and intelligence in construction project management, reducing reliance on subjective expert judgment and enabling data-driven decisions for process optimisation.

According to Kamila and Marzuq (2024), digital tools such as Asana and Trello facilitate project management by providing functionalities to streamline project activities, monitor progress, and enhance collaboration. These tools help organisations manage tasks, schedules, communication, and resources effectively, aligning with the ten project management knowledge areas outlined in PMBOK Version VI[4]. Specifically, Asana is ideal for larger, agile-oriented projects due to its comprehensive features like task management, reporting, team management, and integrations. Trello is better suited for smaller projects, offering simplicity and a user-friendly interface with features like Kanban boards, templates, and integrations. Both tools support project execution by enabling efficient organization, communication, and tracking of tasks (Kamila & Marzuq, 2024). Similarly, Marques and Bernardino (2019) evaluated three tools to facilitate project management: Asana, Odoo[5], and ProjectLibre[6]. They showed that these tools assist in planning, cost management, resource management, and other project-related tasks. Among them, Odoo was identified as the best tool based on the OSSpal methodology (Wasserman et al, 2017), scoring the highest in functionality and overall evaluation.

The analyses of project management tools' effectiveness in supporting knowledge management were performed by Clemente and Domingues (2023). The identified *Targetprocess*[7] as the digital tool that facilitates project management most effectively. It obtained the highest score for knowledge Storage, Sharing, and Application capabilities. Additionally, it excelled in supporting PMBOK artefacts such as "Hierarchy charts" and "Visual Data and Information," making it particularly useful for organising and visualising project-related knowledge. *Confluence*[8] is also highlighted as a strong tool for creating a "knowledge repository," enabling the aggregation, sharing, and editing of project documentation, and it scored highest for knowledge Capture (Clemente & Domingues, 2023).

Challenges in Academic Project Management

Studies have highlighted some recurring challenges in academic project management as it relates to student project execution, including poor time management, ineffective communication, fragmented documentation, and unclear task ownership (Jääskä et al., 2022). These issues are exacerbated by the reliance on manual processes and the lack of digital support tools. The absence of a centralised

[4] https://www.pmi.org/standards/pmbok
[5] https://www.odoo.com/
[6] https://www.projectlibre.com/
[7] https://www.apptio.com/products/targetprocess/
[8] https://www.atlassian.com/software/confluence/use-cases/project-management

project hub often leads to duplicated efforts, missed deadlines, and a lack of accountability within teams. Therefore, the need for a lightweight, easy-to-use, and education-oriented system has become increasingly evident.

Jääskä et al. (2022) identified several challenges in academic project management that affect student project execution. One significant issue is the difficulty in balancing game complexity and challenge, as too much challenge can hinder student attention and engagement. Additionally, not all students are familiar with gaming mechanics, which necessitates proper induction and practice to ensure they can effectively participate in project-based learning. Similarly, Hussein (2021) identified that collaboration challenges arise from priority conflicts among students and uncertainty in project assignments. An inadequate project management structure can lead to unclear expectations and hinder effective communication. Students' interactions and perceptions of each other significantly impact project outcomes, highlighting the importance of a supportive team environment. Addressing these challenges requires a structured approach to project management, emphasising flexibility and adaptability as projects evolve.

Essentially, collaborative learning is a cornerstone of academic development. Research by Allred (2020) and others supports the idea that knowledge construction is most effective when learners engage in social dialogue. Tools that support real-time collaboration, peer discussion, and instructor feedback, such as forums, commenting systems, and evaluation dashboards, have been shown to increase student engagement and project quality (Rapanta & Felton, 2022). The integration of these elements into *TaskHive* is therefore both pedagogically and functionally justified.

Notably, digital transformation in academic institutions has accelerated post-COVID-19, with universities adopting e-learning platforms, virtual labs, and AI-assisted tools to improve academic delivery (Rof et al., 2022). Specifically, digital tools not only improve accessibility but also enhance the quality and consistency of educational experiences. However, the transformation of project-based learning workflows remains underdeveloped. *TaskHive* addresses this gap by aligning project tracking with the broader digital infrastructure of modern educational environments.

This study enhances students' academic project management by providing a tailored digital solution that addresses the unique challenges faced in educational settings. By introducing *TaskHive*, a web-based system specifically designed for student teams, the research offers a structured platform that facilitates efficient task allocation, real-time collaboration, and continuous progress monitoring. Unlike generic project management tools, *TaskHive* integrates features aligned with academic requirements, such as supervisor feedback, document submission, and performance evaluation modules. As a result, students are better equipped to manage time, meet deadlines, and produce higher-quality work, ultimately fostering essential project management skills that are transferable to professional environments.

Methodology

The development of the Web-Based Academic Projects Tracking Management System, also referred to as *TaskHive*, follows a structured and iterative methodology aimed at addressing the unique challenges associated with managing academic projects in higher education. The methodology encompasses both the system's key features and its development lifecycle to ensure functionality, usability, and scalability.

It is designed to streamline project workflows while promoting collaboration, security, and academic accountability. Key features of the system include:

- User Authentication: Secure, role-based access is implemented to ensure that students, supervisors, and administrators interact with the system according to their designated permissions. This maintains data confidentiality and prevents unauthorised access.
- Project Management Module: This core module allows users to create, manage, and track academic projects through structured timelines, task assignments, and progress indicators, ensuring the timely delivery of academic milestones.
- Data Storage and Retrieval: A well-organised database enables structured storage of documents, submissions, and supervisor feedback, facilitating easy retrieval and long-term data management.
- Collaboration Tools: Real-time communication is supported through features such as commenting, feedback sharing, and document exchange, fostering effective interaction between students and their supervisors.
- Automated Reports and Analytics: Supervisors and administrators can access dynamically generated reports that reflect project progress, individual contributions, and workload distribution, thereby improving project oversight and academic support.

The development followed a structured five-phase process to ensure its effectiveness and usability. Initially, requirements analysis was conducted to gather insights from students, supervisors, and administrators, which informed the definition of core functionalities. In the system design phase, the architecture, user interface, and database schema were crafted for intuitive navigation and efficient data flow. The development phase involved building the system using the MERN stack (MongoDB, Express.js, React.js, and Node.js), with additional tools integrated for features like authentication and analytics. Testing and validation were rigorously performed through unit, integration, and user acceptance testing to resolve any issues and ensure system reliability. Finally, the system was deployed to the cloud, accompanied by training and user guides to facilitate smooth adoption by all users.

This methodological approach ensures that *TaskHive* is not only technically sound but also tailored to enhance academic project management, improve team coordination, and support a more structured educational experience.

System Design

The system was developed using the Model-View-Controller (MVC) architectural pattern, as shown in Figure 1. This architecture separates the application into three

interconnected components: the Model, which handles data and business logic; the View, which manages the user interface and presentation; and the Controller, which processes user input and interacts with both the Model and View. This structure promotes scalability, maintainability, and efficient system development.

The system is modular, with functionalities designed to meet the specific needs of academic project tracking. Key interactive interfaces include user and administrator login with role-based access, task and subtask creation, task status updates, user activation by administrators, document and image uploads, and real-time communication between team members. These features are tailored to improve coordination, accountability, and overall project workflow in academic environments.

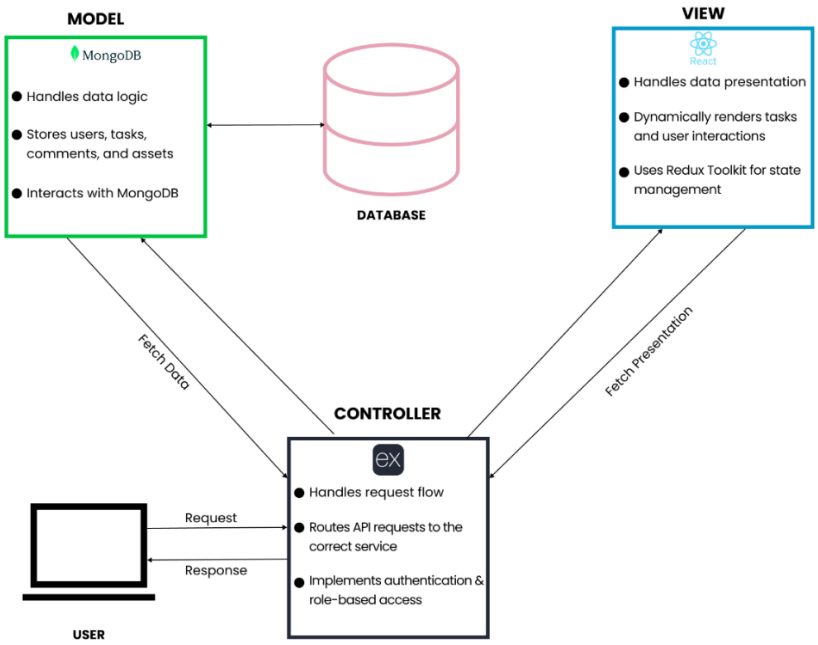


Figure 1: MVC Architectural Pattern, illustrating the separation of concerns across the Model, View, and Controller layers.

Figure 2 presents the sequence diagram, which outlines the chronological flow of interactions among system components and users. It highlights how processes such as login, task assignment, and feedback exchange occur in sequence, providing a clear visual of the system's runtime behaviour. Figure 3 displays the Entity Relationship Diagram (ERD), which represents the logical structure of the database. The ERD defines key entities such as Users, Projects, Tasks, and Submissions, and illustrates how they are interrelated, ensuring consistent data storage and retrieval throughout the system.

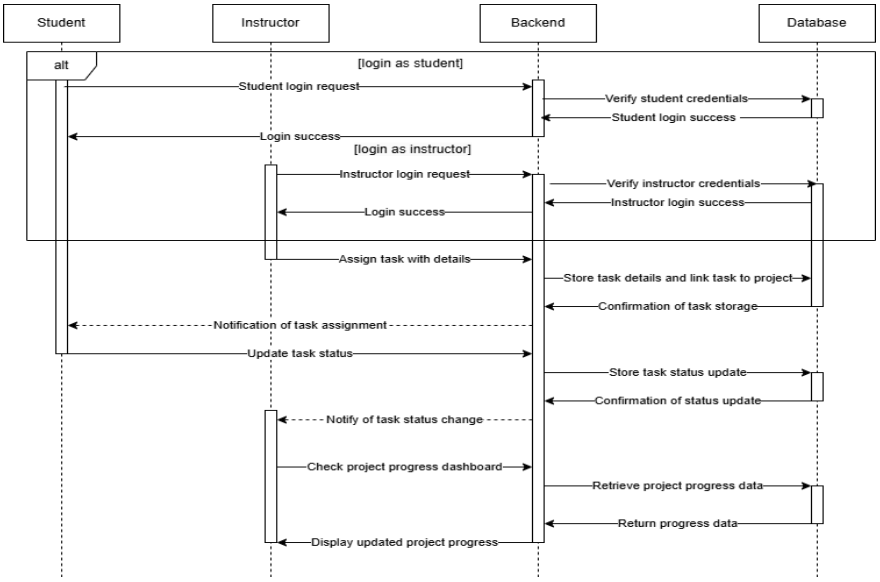


Figure 2: Sequence Diagram showing the flow of interactions between users and system components during typical project management tasks.

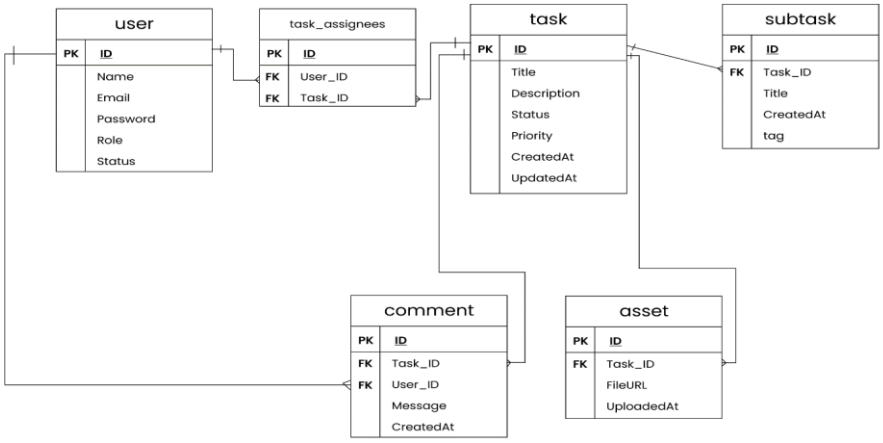


Figure 3: Entity Relationship Diagram (ERD) highlighting the logical structure of the database and the relationships between key entities such as Users, Projects, and Tasks.

Implementation and Testing

The System was developed using the MERN (MongoDB, Express.js, React.js, Node.js) stack, ensuring a scalable, responsive, and full-stack JavaScript solution. The system supports role-based access, with team leads (administrators) responsible for task management and team oversight, while student users engage in project execution and collaboration. Figure 4 displays the user sign-in interface, which provides secure access via email and password authentication.

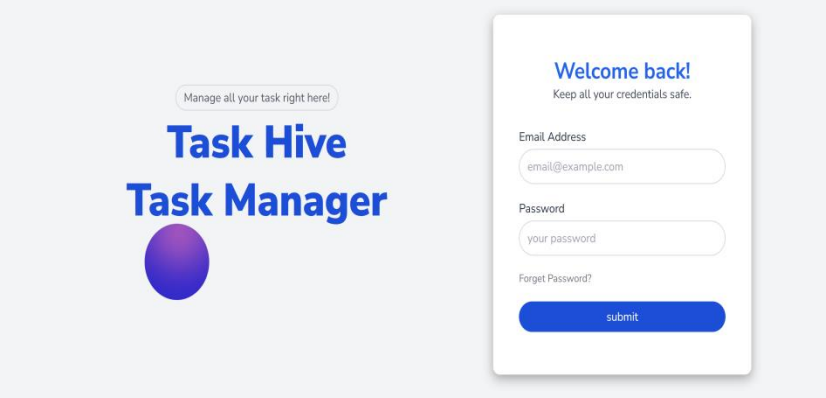


Figure 4: User Sign-In Page

Administrators can efficiently manage projects through the Task Creation and Sub-task Forms, shown in Figures 5 and 6, respectively. These interfaces support the structured assignment of responsibilities, deadlines, and progress tracking.

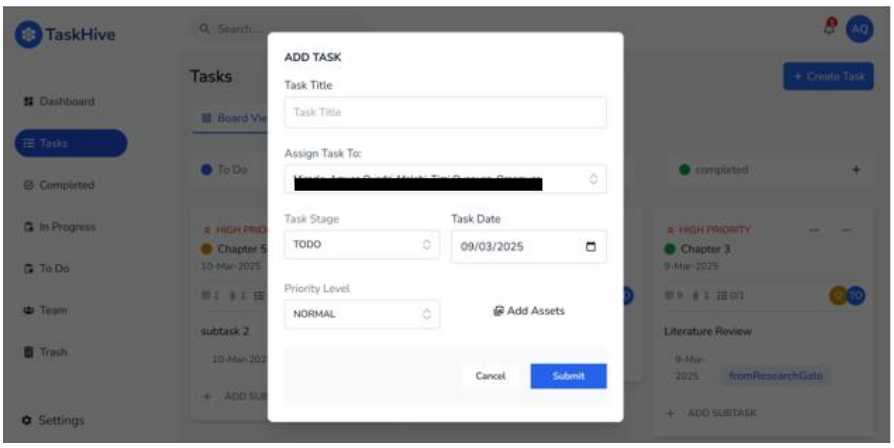


Figure 5: Task Creation Form for Team Lead

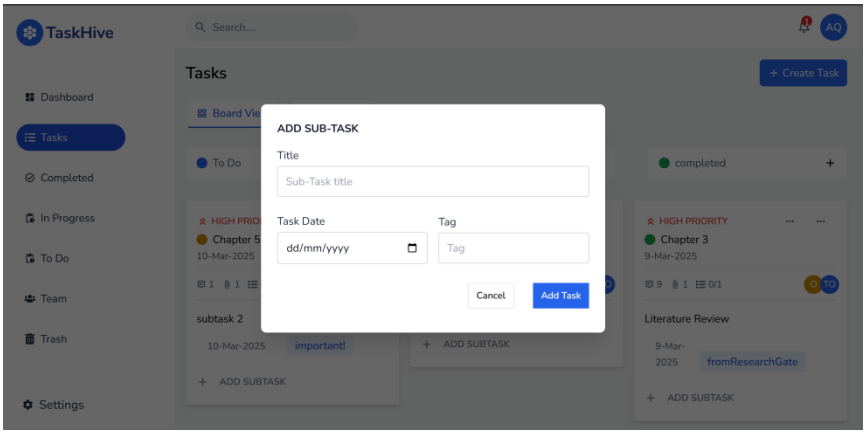


Figure 6: Sub-task Creation Form

Users track progress through status-based task lists, such as assigned and completed tasks (see Figures 7 and 8). These lists enhance transparency and accountability across project teams.

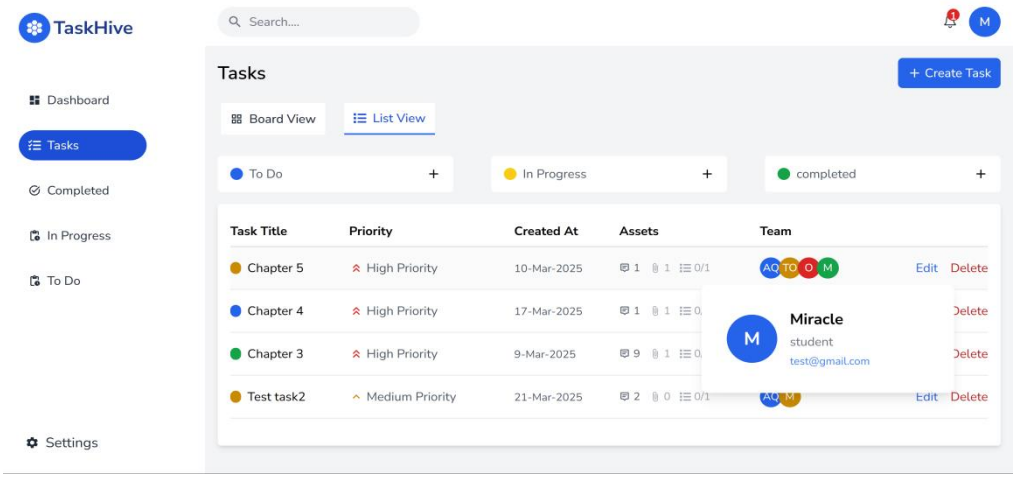


Figure 7: Assigned Tasks List View

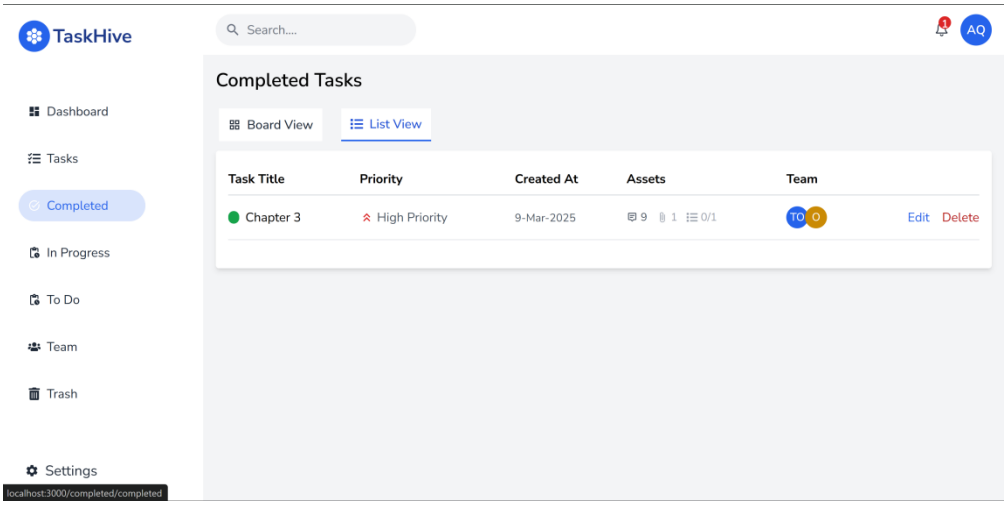


Figure 8: Completed Tasks List View

Real-time communication is supported through a built-in chat interface (Figure 9), enabling direct and immediate collaboration between students and supervisors.

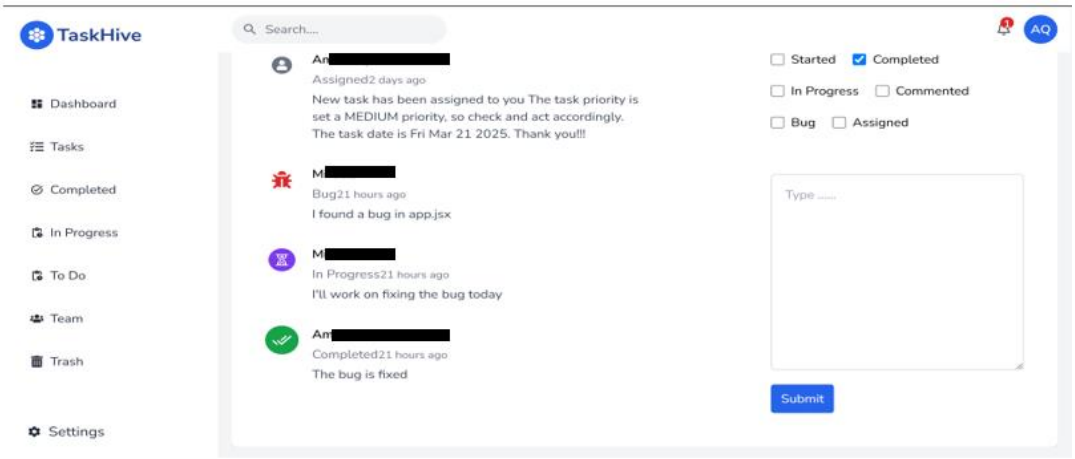


Figure 9: Real-Time Chat Page

Administrative users access a comprehensive dashboard (Figure 10) to manage teams, monitor project metrics, and review overall performance.

Usability Testing

Usability testing was conducted with 15 users comprising students, supervisors, and administrators. Participants were asked to complete a set of tasks (e.g., logging in, assigning tasks, updating statuses, and using the chat feature). The System Usability Scale (SUS) (Brooke, 1996) was used to measure usability, and the system achieved a score of 84.6, indicating *excellent usability*. Key findings include:

- 93% of users found the interface intuitive and easy to navigate.
- 87% reported improved coordination compared to previous project workflows.

- 80% of students indicated they felt more accountable using the system.

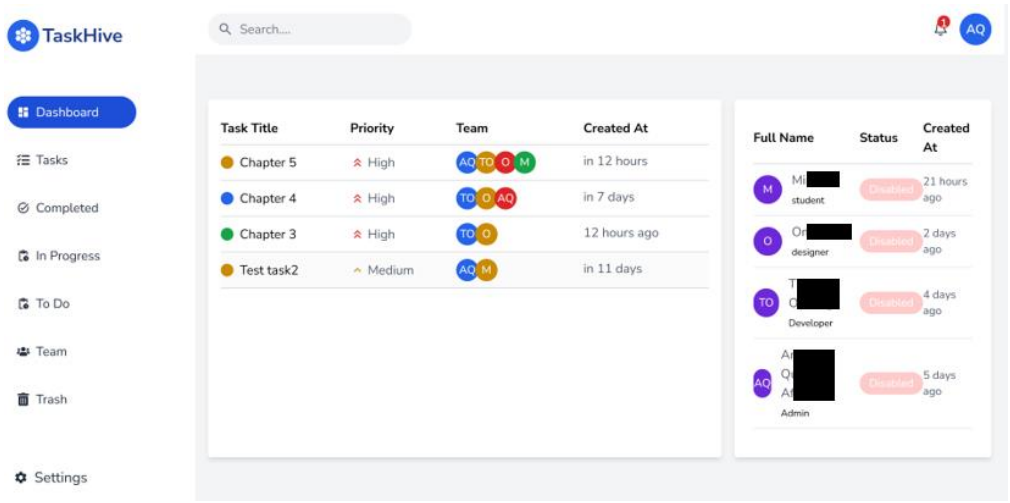


Figure 10: Administrator Dashboard

Performance Metrics

Performance tests were conducted to evaluate system responsiveness and stability under load:

- Average Page Load Time: 1.6 seconds on standard broadband connections.
- Concurrent User Support: Smooth operation with up to 50 concurrent users during stress tests.
- API Response Time: Average of 200ms for core operations such as task creation and updates.
- Uptime: Maintained 99.2% availability during a 14-day test deployment.

These results confirm that the system is both functionally sound and performant under expected usage conditions. Together with positive usability feedback, they demonstrate that the system effectively enhances project coordination and engagement in academic settings.

Discussion

The development and implementation of the Academic Projects Tracking Management System address long-standing inefficiencies in managing academic projects within higher education institutions. Traditional project coordination methods, typically reliant on face-to-face meetings, fragmented communication, and manual task tracking, have proven inadequate in supporting the dynamic, collaborative nature of student-led projects. The system responds to these challenges by offering a structured, digital framework that enhances transparency, accountability, and collaboration.

The integration of core features such as task management, sub-task assignment, real-time messaging, and automated progress tracking aligns with the evolving needs of academic environments. These functionalities mirror best practices in professional project management tools, while remaining intuitive and accessible for student use. Importantly, the system's design, guided by the Model-View-

Controller (MVC) architecture, ensures that scalability, maintainability, and usability are prioritised.

The usability testing results indicate a high level of satisfaction among users, with students reporting improved communication and greater responsibility toward their tasks. Supervisors, on the other hand, highlighted the benefit of centralised oversight, allowing them to monitor individual contributions and provide timely feedback. This shift from disjointed workflows to a unified platform represents a significant improvement in academic project delivery and supervision.

Furthermore, performance testing confirmed the system's robustness, even under moderate load conditions. The system's responsiveness, low API latency, and high uptime reflect the effectiveness of the MERN stack in building reliable, scalable web applications. These attributes are particularly important for academic institutions seeking digital tools that are both cost-effective and user-friendly. By supporting structured workflows, enabling real-time interactions, and providing analytical insights, the system not only improves project outcomes but also contributes to the development of essential digital skills among students. These competencies, such as digital communication, collaborative planning, and time management, are critical for both academic success and future professional engagements.

Compared to existing project management tools, such as Asana, Trello, REDCap, and Targetprocess, *TaskHive* offers a context-specific solution tailored for academic environments. While tools like Asana and Trello are widely adopted for their flexibility and ease of use (Kamila & Marzuq, 2024), they often lack key academic features such as supervisor feedback loops, grading support, and document submission tracking. REDCap, developed for managing clinical research projects, excels in data integrity and coordination but is highly domain-specific and lacks adaptability to educational workflows (Garasević et al., 2024). Similarly, Targetprocess was rated highly for its support of PMBOK artefacts and knowledge management functionalities in professional contexts (Clemente & Domingues, 2023), but it remains too complex and enterprise-oriented for typical student use. In contrast, *TaskHive* incorporates interactive elements grounded in Constructivist Learning Theory (Bada & Olusegun, 2015), enabling peer-to-peer collaboration and student-supervisor communication. It also aligns with the Technology Acceptance Model (Davis, 1989; Marangunić & Granić, 2015) by enhancing perceived usefulness through real-time task monitoring and perceived ease of use via its simple interface. Thus, *TaskHive* addresses a critical gap in existing systems by providing a pedagogically-informed, lightweight, and academically-relevant platform for managing student projects effectively.

Conclusion

This project has successfully demonstrated the development of an Academic Project and Task Management System that addresses key challenges in student project coordination, including role-based access control, real-time task updates, structured collaboration, and progress tracking. The integration of modern web technologies enables a user-friendly interface and responsive system design that supports both

team members and supervisors in managing academic project workflows efficiently. Real-time synchronisation ensures timely information sharing, while the role-based authentication framework upholds data integrity and user accountability.

Despite its strengths, the current implementation focuses on task management within small to medium-sized academic teams and lacks features such as integration with institutional learning management systems (LMS), mobile application support, or automated performance analytics. These limitations suggest avenues for further enhancement to expand system interoperability and scalability.

The system holds practical applications for academic institutions aiming to streamline student project supervision, group collaboration, and progress evaluation. It can also be adapted for research teams and capstone project coordination in higher education. Future research may explore the integration of intelligent recommendation systems for task assignment, AI-driven progress prediction, and broader user testing to evaluate adoption across diverse academic disciplines and institutions.

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